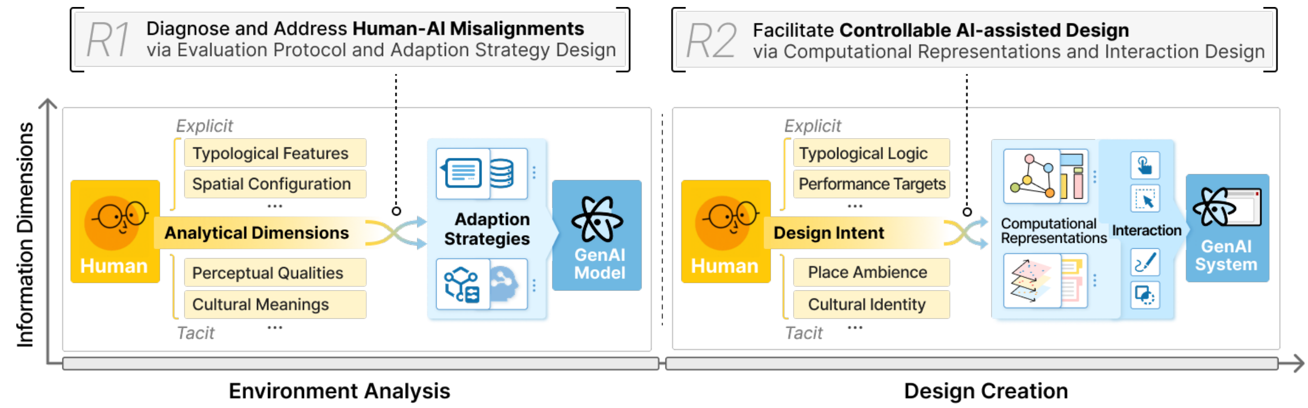
Rong Huang

Ph.D. Candidate
hr3168.github.io
rhuang421@connect.hkust-gz.edu.cn

Research Interests

My research focuses on **Human-AI Interaction in Environmental Design**, intergrating Generative AI and Human-Computer Interaction methodologies in two design stages. For the understanding stage, I **diagnose the misalignment** between human and GenAI, and enable targeted **adaptation strategies to enhance GenAI's capability** to perceive and reason, aligning it with human-centric values. For the creation stage, I **transform design intent into computational representations and design interactive systems** that integrate controllable GenAI into design workflows.



Education

2022 – now • **Ph.D. Candidate, Computational Media and Arts** at *The Hong Kong University of Science and Technology (Guangzhou)*, China, supervised by Prof. Wei Zeng and Prof. Kang Zhang.
Thesis: Human-AI Interaction for Environmental Design: Aligning Generative AI with Design Intent from Understanding to Creation.

2019 – 2022 • **M.Eng., Architecture** at *Tongji University*, China, supervised by Prof. Yu Ye.
Thesis: Economic Benefit Measurement of Street Space Quality and Urban Design Guidance: Based on Multi-source Data and Machine Learning.

2014 – 2019 • **B.Arch., Architecture** at *Hunan University*, China

Publications

1. **Introducing ManyViews: An AI-assisted tool to support citizens' engagement in the design of urban spaces** in *International Journal of Human-Computer Studies*, 2026. DOI: 10.1016/j.ijhcs.2026.103796.
Rong Huang, Yihan Hou, Mela Bettega, Kang Zhang, and Wei Zeng* *(SCI Q1)*

2. **Fidelity-driven data augmentation for multimodal large language model on architectural heritage interpretation** in *npj Heritage Science*, 2026. DOI: 10.1038/s40494-026-02446-2.
Rong Huang, Hai-chuan Lin, and Wei Zeng* *(SCI Q1)*

3. **PlantoGraphy: Incoporating iterative design process into generative artificial intelligence for landscape rendering** in *Proceedings of ACM CHI Conference on Human Factors in Computing Systems*, 2024. DOI: 10.1145/3613904.3642824.
Rong Huang, Hai-chuan Lin, Chuanzhang Chen, Kang Zhang, and Wei Zeng* *(CCF-A, Citation: 60)*

4. **SceneWeaver: A multi-agent collaborative system for 3D scene creation in video games** in *Proceedings of the International Symposium on Visual Information Communication and Interaction*, 2025. DOI: 10.1145/3769534.3769540.
Rong Huang, Chenxi Ruan, Bingchuan Jiang, and Wei Zeng*

5. **Unified visual comparison framework for human and AI paintings using neural embeddings and computational aesthetics** in *IEEE Computer Graphics & Applications*, 2025. DOI: 10.1109/MCG.2025.3555122.
Yilin Ye, Rong Huang, Kang Zhang, and Wei Zeng* *(SCI Q3)*

6. **HeritageExplorer: Interactive visualization and dialogue system for multi-modal architectural heritage exploration** in *Proceedings of the International Symposium on Visual Information Communication and Interaction*, 2025. DOI: 10.1145/3769534.3769571.
Yusong Wang, Yihan Hou, Rong Huang, and Wei Zeng*

7 **VISAtlas: An image-based exploration and query system for large visualization collections via neural image embedding** in *IEEE Transactions on Visualization and Computer Graphics*, 2024. DOI: 10.1109/tvcg.2022.3229023.
Yilin Ye, Rong Huang, and Wei Zeng* *(SCI Q1, Citation: 46)*

8 **Is it the end? guidelines for cinematic endings in data videos** in *Proceedings of ACM CHI Conference on Human Factors in Computing Systems*, 2023. DOI: 10.1145/3544548.3580701.
Xian Xu, Aoyu Wu, Leni Yang, Zheng Wei, Rong Huang, Yip David, and Huamin Qu* *(CCF-A, Citation: 17)*

9 **A data-informed analytical approach to human-scale greenway planning: Integrating multi-sourced urban data with machine learning algorithms** in *Urban Forestry & Urban Greening*, 2020. DOI: 10.1016/j.ufug.2020.126871.
Ziyi Tang, Yu Ye, Zhidian Jiang, Chaowei Fu, Rong Huang, and Dong Yao *(SCI Q1, Citation: 89)*

Research Projects

2025.01 – 2027.12 • **Guangxi Key Research and Development Program:** "Key Technologies for Trustworthy Intelligent Integration of Guangxi Traditional Architecture Based on Multimodal Data" (PI: Prof. Wei Zeng)
Funding: ¥600K out of ¥1.5M.
Role: Project Coordinator
- Contribute to the grant proposal, defining the core technical roadmap and the project timeline.
- Lead and manage the research team, and responsible for the core technical development and key research outputs.
- Prepare progress reports and technical documentation.

2020.01 – 2022.06 • **National Natural Science Foundation of China:** "Measuring public space quality: An evaluation model and its design support based on multi-sourced urban data and deep learning algorithms" (PI: Prof. Yu Ye)
Role: Student Researcher
- Multi-sourced urban dataset construction using Python and GIS.
- Statistical analyses to support the development of the evaluation model.

2020.01 – 2021.12 • **National Natural Science Foundation of China:** "Walkability of street interfaces: A fine-scale measurement and design control methods" (PI: Prof. Yu Ye)
Role: Student Researcher
- Performed data statistical analysis using Python.
- Managed the visualization and layout of research findings for publication.

Work Experience

2019 – 2022 • **Research Assistant,** the Computational Urban Design Research Centre (PI: Dr. Yu Ye), Joint Laboratory for International Cooperatuon on Eco-Urban Design (Tongji University), Ministry of Education, China.

2021.06 – 2021.09 • **Urban Design Intern,** Skidmore, Owings & Merrill LLP (SOM), Urban Design Department, Shanghai Office, China.

2018.06 – 2018.09 • **Architecture Design Intern,** URBANUS, architecture and urban design practice, China.

Awards

2019.12 • **First Prize,** Shanghai Urban Design Challenge (award ¥100K about $14,000). The best team among 105 teams from different countries.

2016.06 • **Research Prize,** Architecture and Urban Design: Case Study House in USA (award ¥20K about $3,000). Only the top 1% students were prized among 500 peers.

Skills

Design Tools • Interactive Prototyping (Figma), 3D Modeling & Animation (Rhino, SketchUp, Cinema4D, Unity, Unreal Engine), Adobe Suites

Analysis & Programming • Python, ArcGIS, SPSS, P5.js, PyTorch

HCI Methods • User-Centered Design, Qualitative Methods (Thematic Analysis, Interviews), Quantitative Methods (Statistical Analysis), Prototyping

Languages • Mandarin (Native), English (Professional Proficiency), Cantonese, Hakka